Team HAP

Leveraging OpenAI for Business Analytics

Project Plan

Michigan State University

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HAP Sponsors

Steve Neubecker

Angela Endres

Michigan State Capstone Team

Evan Bean

Lily Hami

Ziqi Liu

Brendan Murphy

Sukruth Rao

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Executive Summary

HAP, which stands for Health Alliance Plan, is a leading not-for-profit health insurance company headquartered in Michigan. With a rich history dating back to 1956, HAP has continually evolved to meet the dynamic healthcare needs of its diverse customer base. Through a comprehensive array of health insurance plans, HAP provides individuals, families, and businesses with access to high-quality healthcare services, fostering a culture of wellness and vitality across the state.

What sets HAP apart is its deep-rooted local presence and community-driven approach. As a Michigan-based health insurance company, HAP understands the region's unique healthcare challenges and actively contributes to its betterment through partnerships, sponsorships, and outreach programs. HAP's commitment to innovation is evident in its embrace of cutting-edge technology and telehealth solutions, ensuring that members have convenient and accessible healthcare options. With a strong focus on preventive care, HAP empowers its members to proactively manage their health, ultimately leading to healthier, happier lives. In summary, HAP is not just a health insurance provider; it's a trusted partner in promoting the health and well-being of the Michigan community.

In alignment with HAP's history of evolving to meet customer needs, Team HAP is constructing a web application to showcase how artificial intelligence and machine learning can be used to improve business processes. This web application will enable HAP to better understand their customer base and allow them to consistently offer plans that best align with customer needs.

Functional Specifications

Finding the right healthcare plan can be a stressful and time-consuming process. In today's world, the general population is aging, leading more people to search for healthcare plans, specifically Medicare. This often results in confusion and questions when individuals are trying to determine the best path for their future. The primary goal of the features within this web application is to streamline the process for healthcare customers, making it easier for them to receive the medical care they need.

This project has three main objectives:

- Information Retrieval: Help HAP customers quickly access answers available on hap.org.
- Plan Selection Assistance: Aid customers in choosing the most suitable health insurance plan.
- Enhanced Customer Service: Improve HAP customer service by understanding the reasons behind customer calls.

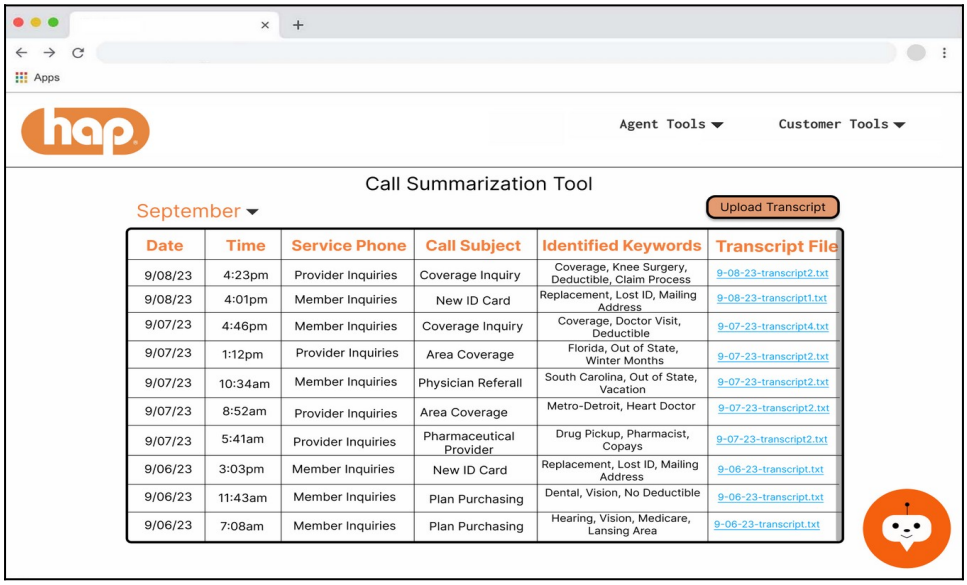
The software behind these features will analyze hap.org content to provide customers with the answers they are looking for more quickly and efficiently. This will offer an alternative route within the website to expedite the shopping experience and alleviate the headaches that come with trying to find answers to this complex topic. The Plan Selector tool will further simplify things by allowing customers to navigate this process on their own without needing to interact with customer service employees. Customers will be asked a series of detailed questions that will help HAP learn more about their needs and preferences to find the best plan for them and their future. By studying call transcription data, we can train our new software to anticipate the needs of new insurance customers even before interacting with them. This project will combine all these exciting new features to enhance the customer experience and simplify a rather complex procedure.

Design Specifications

Overview

The web application is designed to showcase the potential of OpenAI and Machine Learning in addressing user experience challenges. It will be providing solutions related to shopping, search queries, and assisting employees in identifying common reasons for customer service calls. It will feature a user-friendly interface for viewing call transcript summaries, a comprehensive shopping tool, and an informative chatbot for answering questions, all aimed at demonstrating these capabilities to our client.

Call Summarization Tool



Date	Time	Service Phone	Call Subject	Identified Keywords	Transcript File
9/08/23	4:23pm	Provider Inquiries	Coverage Inquiry	Coverage, Knee Surgery, Deductible, Claim Process	9-08-23-transcript2.txt
9/08/23	4:01pm	Member Inquiries	New ID Card	Replacement, Lost ID, Mailing Address	9-08-23-transcript1.txt
9/07/23	4:46pm	Member Inquiries	Coverage Inquiry	Coverage, Doctor Visit, Deductible	9-07-23-transcript4.txt
9/07/23	1:12pm	Provider Inquiries	Area Coverage	Florida, Out of State, Winter Months	9-07-23-transcript2.txt
9/07/23	10:34am	Member Inquiries	Physician Referall	South Carolina, Out of State, Vacation	9-07-23-transcript2.txt
9/07/23	8:52am	Provider Inquiries	Area Coverage	Metro-Detroit, Heart Doctor	9-07-23-transcript2.txt
9/07/23	5:41am	Provider Inquiries	Pharmaceutical Provider	Drug Pickup, Pharmacist, Copays	9-07-23-transcript2.txt
9/06/23	3:03pm	Member Inquiries	New ID Card	Replacement, Lost ID, Mailing Address	9-06-23-transcript.txt
9/06/23	11:43am	Member Inquiries	Plan Purchasing	Dental, Vision, No Deductible	9-06-23-transcript.txt
9/06/23	7:08am	Member Inquiries	Plan Purchasing	Hearing, Vision, Medicare, Lansing Area	9-06-23-transcript.txt

Figure 1: Call Summarization Tool

Figure 1 displays the call transcript summarization page. The user can upload a text file containing a transcript from a customer service call by clicking on the “Upload Transcript” button in the upper right corner of the page and view details about the call in the table. Starting from the left side of the table, the user will be able to view the date and time of the call, whether it was a call from a customer or provider, the subject of the call, as well as any keywords identified. The user can also access transcripts from different months by using the dropdown menu in the upper left corner of the page.

Call Transcript Data Summarization

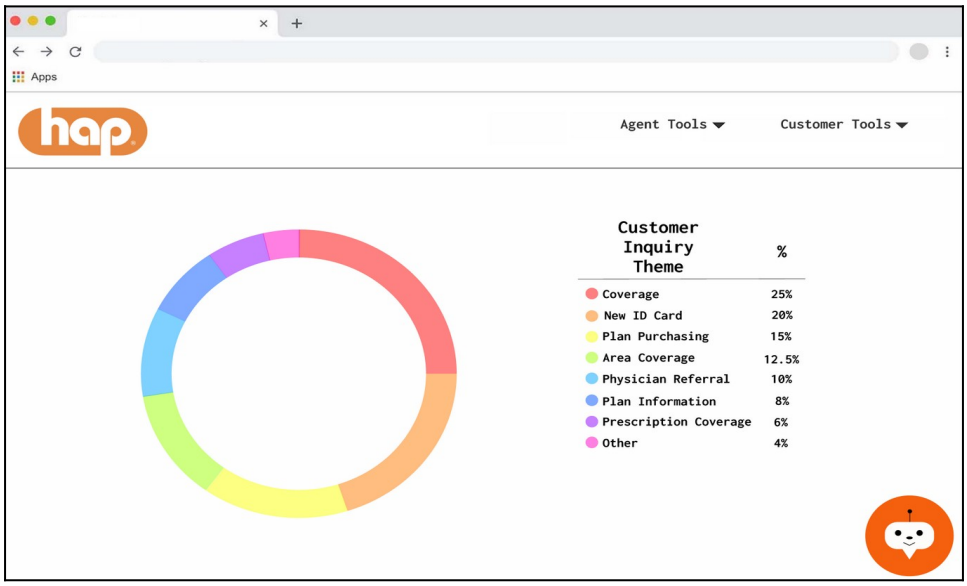


Figure 2: Call Transcript Data Summary Tool

Figure 2 illustrates how the application will enable users to rapidly visualize the categorization of call transcripts. Access to this data page can be found using the navigation bar under 'Agent Tools.' The customer inquiry themes displayed in the chart correspond to the 'Call Subject' column shown in Figure 1.

Chatbot Tool

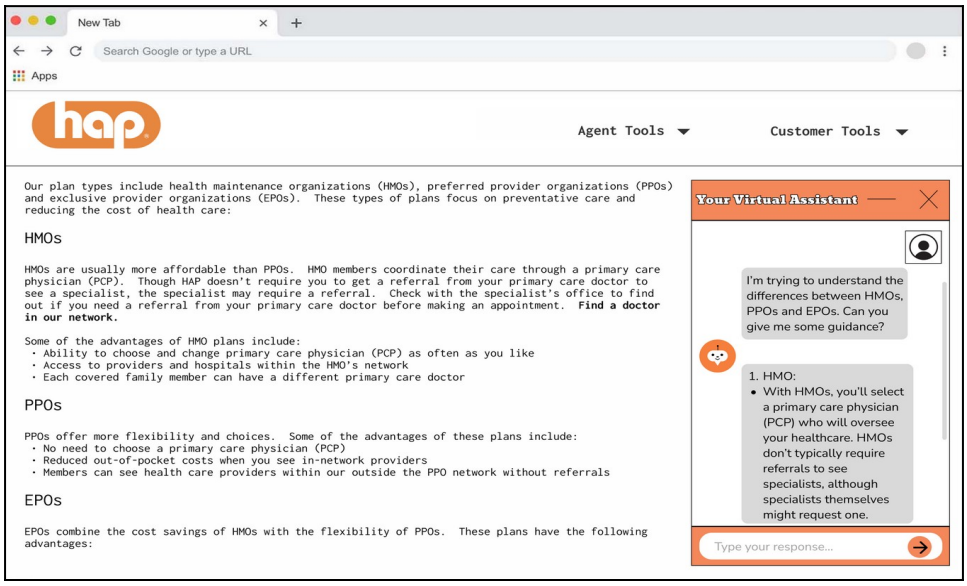


Figure 3: Chatbot Feature for a Clarification Question

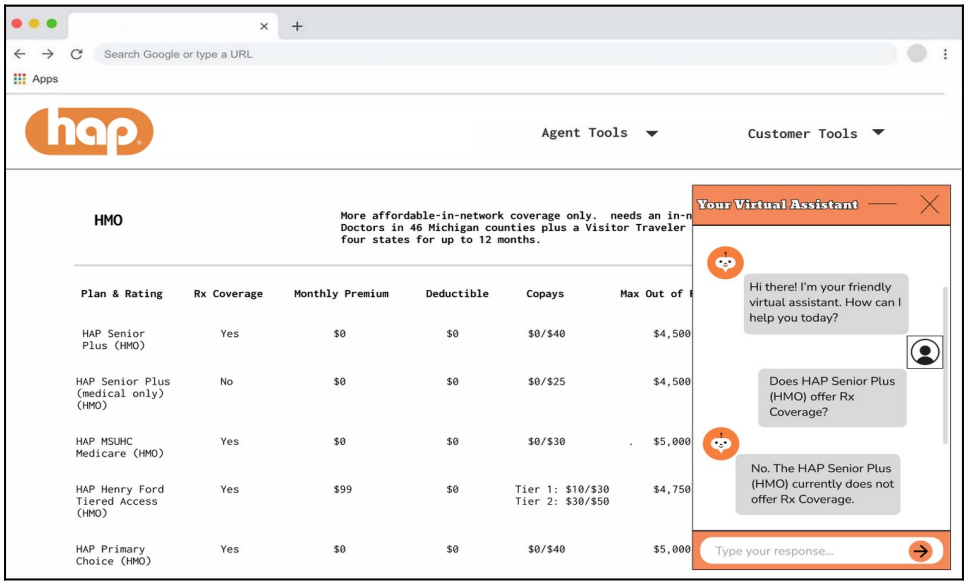


Figure 4: Chatbot Feature for a Coverage Question

Figures 3 and 4 show how utilizing a chatbot will enable hap.org users to quickly find answers to questions that would otherwise require more time to locate. In Figure 3, we observe an example of a customer who is confused about the differences between HMOs, PPOs, and EPOs. By opening the chatbot and sending a quick question, the user can receive guidance on what distinguishes these plans. In Figure 4, we see a user who wants to determine if a specific plan offers the coverage they need. Instead of searching through plans and their respective documents, the chatbot can promptly provide an answer to their question.

Plan Selector Tool

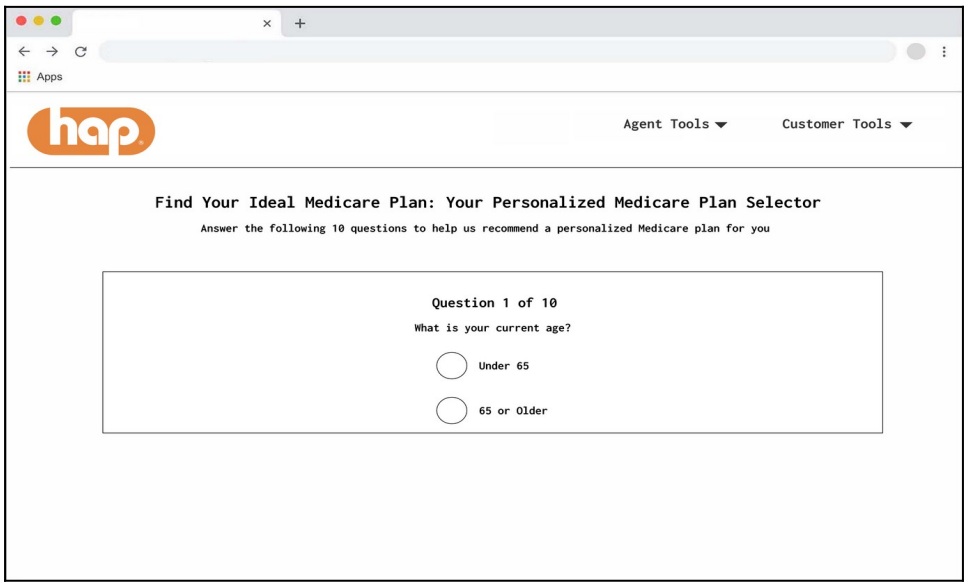


Figure 5: Plan Selector Tool Quiz – Option Selection Interface

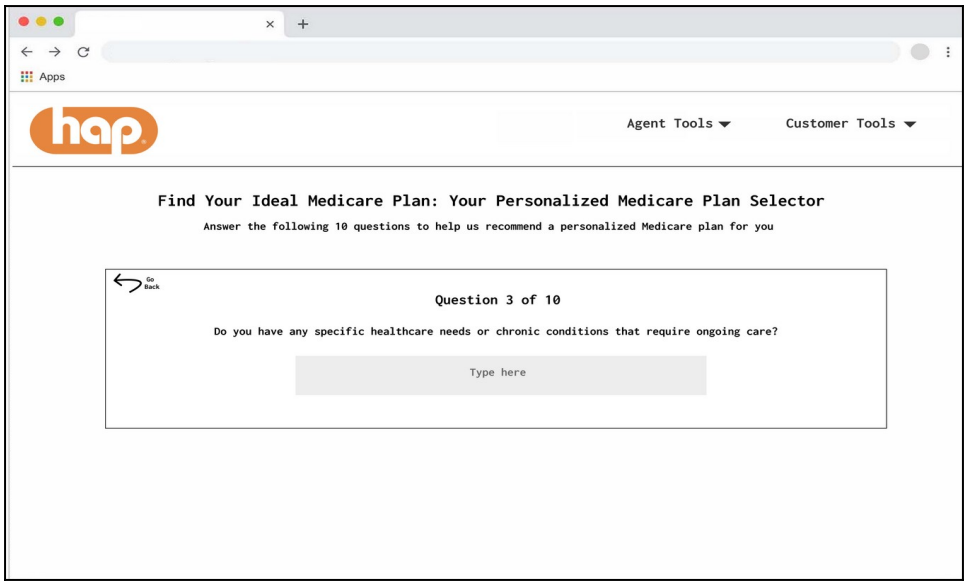


Figure 6: Plan Selector Tool Quiz—Text Input Interface

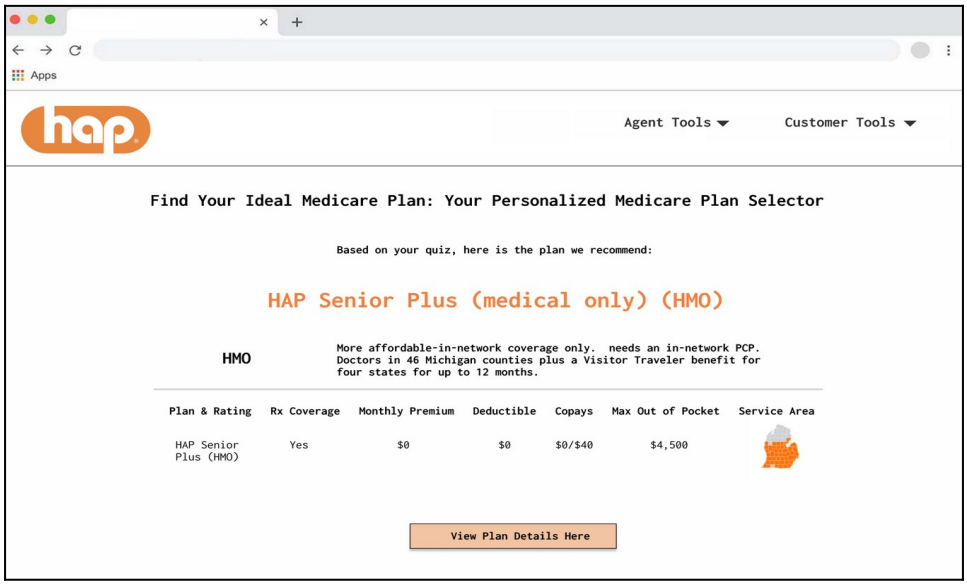


Figure 7: Plan Selector Tool Quiz Results

Figures 5, 6, and 7 illustrate the user flow of a Medicare plan selector tool. Users will engage in a ten-question quiz in which they can either select an option (as depicted in Figure 5) or input an answer (as shown in Figure 6). After advancing past the first question, users will have the option to revisit a previous question, preventing them from restarting the quiz. They can do this by clicking on the back arrow located in the upper left corner of the question box in Figure 6. Figure 7 provides a glimpse of what users will see upon completing the quiz, where a Medicare plan will be suggested based on their answers. The display will feature the plan's name and additional details, such as its rating, inclusion of prescription coverage, monthly premium cost, deductible, copay amounts, maximum out-of-pocket expenses, and a service area map. Furthermore, users will have the option to click a button at the bottom of the page, which will redirect them to an Hap.org page with more information about the specific plan. This tool is designed for casual shoppers who prefer not to engage with the chatbot or contact customer service.

Technical Specifications

System Architecture Diagram

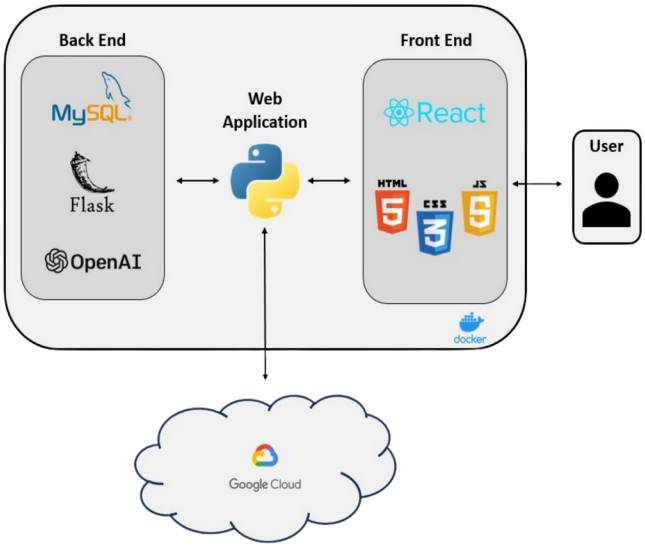


Figure 8: System Architecture

The web application under development features a modern system architecture as shown in Figure 8. On the front-end, it utilizes React, HTML, CSS, and JavaScript. React is employed for creating dynamic and responsive user interfaces, while HTML and CSS structure and style the web pages. JavaScript enhances interactivity and user experiences.

For the back-end, Flask, a lightweight and versatile Python web framework, handles HTTP requests, routing, and communication with the MySQL database and OpenAI services. MySQL serves as the relational database management system, ensuring efficient data storage, retrieval, and management. OpenAI integration on the back-end empowers the application with natural language processing and AI capabilities, enabling features such as chatbots and text generation.

To ensure consistent and reliable deployment, the application is containerized using Docker. This approach simplifies packaging, facilitating deployment, scaling, and maintenance. Google Cloud Platform (GCP) is used as the hosting and deployment environment.
The web application features a well-structured combination of front-end and back-end technologies, all containerized within Docker for seamless deployment on the Google Cloud

Platform. This stack offers the flexibility, scalability, and reliability required for creating a modern and efficient web application with features powered by OpenAI's AI capabilities.

System Components

React
A JavaScript library that is used to build user interfaces for web applications and allows developers to create dynamic and responsive front-end components. React efficiently updates the user interface in response to changes in data making it a powerful tool for creating interactive and visually appealing web applications.

HTML
A standard markup language for defining the structure and elements within a web application. It provides a skeleton that is then enhanced by CSS and JavaScript.

CSS
A stylesheet language used in web development to define the presentation and visual styling of HTML documents. It allows developers to control layout, colors, fonts, spacing, and other attributes to ensure a web page is visually appealing and consistent across different browsers and devices.

JavaScript
A high-level programming language used in web development to add functionality and interactivity to web pages. It allows developers to create dynamic content, interact with APIs, and handle user input to create modern and interactive web applications.

Docker
A platform used for developing, shipping, and running applications in a lightweight isolated container. Docker simplifies the development and deployment process to make managing complex applications easier.

Python Web Application
A software application that uses the Python programming language for back-end logic.

MySQL
An open-source relational database management system used to store and manage structured data. It is known for its reliability, scalability, and performance and is commonly used as the database back-end for web applications.

Flask

A lightweight and flexible Python web framework used when building web applications. It is known for its simplicity and extensibility.

OpenAI
An artificial intelligence research lab and technology company that focuses on developing advanced AI systems and tools. They have produced many AI models and technologies including GPT-3, GPT-4, other language models, and reinforcement learning agents for a multitude of applications. OpenAI's mission is to ensure that artificial general intelligence benefits all of humanity.

OpenAI API
A programmatic interface by OpenAI that allows developers to integrate OpenAI's language models and AI capabilities into their own services and applications. Developers can use the API to interact with pre-trained models to allow them to harness the power of OpenAI's technology within their own projects.

OpenAI Embeddings
Refers to the representations of text data generated by OpenAI's language models. These embeddings capture context and semantic meaning to be used for natural language processing tasks and can also be used for tasks such as search and text classification.

Vector Database
A database system designed to store and manage data within the web application.

Testing Plan

Selenium (Front-End Testing)
Selenium proves to be a valuable tool for our project as it allows us to replicate genuine user interactions with the website. By using Selenium, we can validate the functionality and responsiveness of key elements like buttons, forms, menus, and links, ensuring the correctness and usability of the website's user interface.

Pytest (Back-End Testing)
Pytest is a popular and widely used testing framework for Python. It provides a comprehensive and flexible platform for writing and executing test cases for Python applications. Pytest is known for its simplicity, scalability, and ease of use, making it a preferred choice for many developers. It can be used to test Python code and web applications that utilize Flask, API

endpoints, and data processing functions. Since our team is utilizing these technologies in our web application, this framework for back-end testing will be most effective.

Risk Analysis

Risk 1: Inaccurate Summarization of Call Content by OpenAI

Risk Level: Medium

Description: When summarizing a call transcript, there is a possibility that OpenAI may provide an incorrect summary of the call's content. For instance, if a call initially pertains to a coverage inquiry but is mistakenly categorized as a lost ID card inquiry by the AI, it could negatively impact the reliability of our product.

Mitigation: To address this risk, we will subject a varied set of call transcripts to OpenAI's processing to assess the model's reliability. This evaluation will help determine whether processing a transcript multiple times can yield more precise and dependable results.

Risk 2: Ensuring Compliance with Protected Health Information (PHI) in Call Transcript Processing

Risk Level: Low

Description: Call transcripts may include sensitive Protected Health Information (PHI), encompassing health histories and personal data. It is imperative to uphold privacy while dealing with these call transcripts.

Mitigation: We will address this concern by developing our product using simulated call transcripts created by ChatGPT. Our client has indicated their intention to use our product as a means of testing the technology's capabilities. This approach will allow us to avoid violating PHI laws.

Risk 3: Cost Implications of OpenAI API Usage due to Token Requirements for Integration

Risk Level: Low

Description: OpenAI's API generates text and consumes tokens for every X characters, the specific number varying based on the model in use. While the cost per individual character is minimal, these expenses can accumulate during the development phase.

Mitigation: To address this issue, we will engage in open communication with the client to establish a clear budget for project development. We will then optimize our token usage while staying within the defined budgetary constraints. This ensures responsible management of financial resources allocated to us.

Risk 4: Dependency on a Third-Party API as the Core Functionality of Our Web Application

Risk Level: Medium

Description: The primary functionality of our web application heavily relies on OpenAI. However, it's essential to acknowledge that the availability of the OpenAI service cannot be guaranteed at all times.

Mitigation: In the event that the OpenAI service is temporarily unavailable, we will redirect users to an alternative resource or give them a default response, clearly indicating the unavailability of the service. This ensures a smoother user experience during such situations.